THE COORDINATES OF A QUARTIC L-VALUE: A FORMALIZED SURVEY OF THE CLASSICAL FACTS

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ABSTRACT. For primes $p > 5$ with $p \equiv 5 \pmod 8$, the quartic generalized Bernoulli number $\beta = B_{1,\chi_4}$ is, after a choice of quartic character, a Gaussian integer $k+it$. This survey collects in one place the classical facts about these coordinates that are otherwise spread across papers from 1976 to 2026: why order four is the first level at which an odd non-real character occurs at prime conductor; what the trace and the relative class number already determine (k and $|t|$); how the normalization $\chi_4(2) = i$ fixes the sign of t and agrees with the quartic Jacobi-sum normalization and with Hudson's quarter-range coordinates; and the fixed-base digit identity together with what it does not reach. A section on the higher 2-power levels records, for characters of order 2^a modulo primes with $v_2(p-1) = a$, that the coset differences vanish below the top level, that the coordinates of $B_{1,\chi}$ in the power basis are odd integers, and that their first coordinate is the trace of Murty and Thangadurai. Nothing in this survey is new; every statement is attributed to its source, and the one statement we did not find in print, the parity of the coordinates at levels $2^a \geq 8$, is the argument of the level-four case verbatim. The elementary core (Propositions 1.1, 3.1, 3.2, 3.4 except one remark, 3.5, 4.1 and 5.1 to 5.3, Definition 3.3, and the conjugation and reconstruction statements) is formally verified in Lean 4 with Mathlib; Section 7 lists what is verified and what is quoted.

1. The classical boundary

In 1839 Dirichlet proved that for a prime $p \equiv 3 \pmod 4$, $p > 3$, the interval $(0, p/2)$ contains more quadratic residues than non-residues, and that the excess is $(2 - \left(\frac{2}{p}\right))h(-p)$, that is, $h(-p)$ when $p \equiv 7 \pmod 8$ and $3h(-p)$ when $p \equiv 3 \pmod 8$ [1, (1.1)]. The programme that grew out of this asks, for each interval and each class of primes, whether the corresponding character sum has constant sign and what class number measures it. Berndt's 1976 survey [1] is the systematic treatment: it works through intervals of length $k/3$, $k/5$, $k/6$, $k/8$, $k/10$, $k/12$, $k/16$ and $k/24$, expresses the sums in terms of Dirichlet L-values at $s = 1$, translates the real cases into class numbers, and deduces congruences.

Twice in that paper Berndt identifies the obstruction. At the sixteenth range [1, §10] the individual sums involve L-functions of complex characters, so his methods yield no conclusion about their signs; the theorem that follows concerns paired sums. In the closing section [1, §14] he states the general point: every such interval sum can be written as a linear combination of L-functions at $s = 1$, but when non-real characters enter the sign argument stops. At the fifth range [1, §5] he names quartic characters as the obstruction to a positivity conclusion.

The same boundary appears in the digit literature. Girstmair [9, Rem. 5] treats the frequency of the digit 0 in the decimal period of $1/p$ when 10 has order $(p-1)/2$. For $p \equiv 3 \pmod 4$ the answer is a combination of $h(-p)$ and $h(-5p)$. For $p \equiv 1 \pmod 4$ the evaluation supplied by [21, (6)] involves a first generalized Bernoulli number attached to a character of order four, and Girstmair reports no class-number interpretation of the individual complex value.

Classical methods do not stop at this boundary. They continue to control Galois-invariant combinations, namely traces, norms, determinants and class-number formulas. What they do

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not give is an ordered sign for the individual quartic value, which does not collapse to a signed scalar class number in the quadratic manner. What is left over is the ordered pair of coordinates after the conjugation ambiguity has been fixed.

Proposition 1.1 (the first non-real odd level)**.** *Let χ be a Dirichlet character modulo a prime p and suppose that χ has exact order $q > 1$. For $q = 2$ the character is real. For odd q (in particular $q = 3$) every such character is even. At $q = 4$ the character is odd and non-real precisely when $p \equiv 5 \pmod 8$. Thus order four is the first character order at which an odd non-real character can occur at prime conductor.*

Proof. If q is odd, then $\chi(-1)$ has order dividing both q and 2, hence $\chi(-1) = 1$. A character of order two takes only the values 0 and ± 1. If χ_4 has order four, choose a generator g of $(\mathbb{Z}/p\mathbb{Z})^\times$ with $\chi_4(g) = i$. Since $-1 = g^{(p-1)/2}$, $\chi_4(-1) = i^{(p-1)/2} = (-1)^{(p-1)/4}$. Hence χ_4 is odd exactly when $(p-1)/4$ is odd, equivalently $p \equiv 5 \pmod 8$. In that case χ_4 and $\overline{\chi_4}$ are the two odd characters of order four and are non-real. □

Lean: `apply_neg_one_of_odd_orderOf`, `apply_of_orderOf_two`, `quartic_apply_neg_one`, `quartic_apply_neg_one_of_one`.

This elementary boundary is the one visible in Girstmair's coset-sum method. His sign-vector argument [7] requires a nontrivial quadratic character among the odd characters, a condition that holds in the real cases but fails at index four; his 2026 survey [11] accordingly treats $q = 2$, 6 and 10 rather than $q = 4$.

2. What classical theory determines

Throughout, p is a prime with $p > 5$ and $p \equiv 5 \pmod 8$; write $\mathcal{P}^*$ for this set. Let χ_4 be one of the two primitive quartic characters modulo p and set

$$\beta = B_{1,\chi_4} = \frac{1}{p} \sum_{1 \le n \le p-1} \chi_4(n)\, n.$$

As shown in Proposition 3.1, β lies in $\mathbb{Z}[i]$; write $\beta = k + it$. Let $\langle x \rangle$ denote the least nonnegative residue of x modulo p, let H be the subgroup of fourth powers modulo p, and, for $j = 0, 1, 2, 3$, put

$$C_j = \{n \in (\mathbb{Z}/p\mathbb{Z})^\times : \chi_4(n) = i^j\}, \qquad S_j = \frac{1}{p} \sum_{n \in C_j} \langle n \rangle.$$

Thus $C_0 = H$. Up to the cyclic indexing determined by χ_4, the S_j are the classical coset sums and, in Girstmair's terminology [7], the components of the b-deviation vector at $b = p + 1$.

The L-value normalization. For every nonprincipal character χ, $L(0, \chi) = -B_{1,\chi}$. If χ is primitive and odd of conductor p, the functional equation gives

$$L(1, \overline{\chi}) = \frac{\pi i\, \tau(\overline{\chi})}{p}\, B_{1,\chi},$$

where $\tau(\overline{\chi})$ is the Gauss sum. Thus $B_{1,\chi}$ is the algebraic normalization of the corresponding value at $s = 1$, and the coordinates in this survey are the coordinates of that normalized quartic L-value.

2.1. **Trace, norm and the relative class number.** Let K_p be the imaginary cyclic quartic subfield of $\mathbb{Q}(\zeta_p)$, and let $h^*(p) = h(K_p)/h(\mathbb{Q}(\sqrt{p}))$ be its relative class number.

Proposition 2.1 (the classical data)**.** *For $p \in \mathcal{P}^*$,*

$$\mathrm{Tr}(\beta) = 2k, \qquad \mathrm{N}(\beta) = k^2 + t^2 = 2h^*(p). \tag{1}$$

Consequently the trace and the relative class number determine k and $|t|$, and the only ambiguity in the individual quartic value is the conjugation $t \mapsto -t$.

Proof. The trace identity is formal from $\beta = k + it$; its occurrence as a classical digit invariant is [18, Thm. 1], restated at the 2-power levels in [10, Thm. 1] (see Proposition 5.3). Also $\beta = (S_0 - S_2) + i(S_1 - S_3)$. Hudson and Williams [14], with proof and discussion in Hudson [15, (2.3)], give $h^*(p) = \frac{1}{2}\big((S_2 - S_0)^2 + (S_3 - S_1)^2\big)$. Substitution gives $\mathrm{N}(\beta) = 2h^*(p)$. Equivalently, this is the analytic class number formula $h^* = Qw\prod(-\frac{1}{2}B_{1,\chi})$ over the two odd characters, with $Q = 1$ and $w = 2$. □

Lean: the trace, as `trace_eq`. The identification of the norm with $2h^*(p)$ needs class numbers of number fields and is quoted, not formalized.

The trace fixes the real coordinate and the relative class number fixes the absolute value of the imaginary coordinate, so classical theory determines the quartic value up to complex conjugation. The role of the pin in Section 3 is to choose between the two conjugates.

The same norm structure persists in higher degree. Girstmair [4], restated precisely by Kučera [16], expresses relative class numbers of imaginary abelian fields of prime-power conductor as group determinants in the coset sums, with explicit formulas up to degree ten. On the trace side, Murty and Thangadurai [18, Thm. 1] show that for $r \mid p - 1$, with the relevant characters odd, a digit sum of $1/p$ is an elementary term plus a trace of first generalized Bernoulli numbers. At $r = 4$ that trace is $2k$; at the higher 2-power levels it is Proposition 5.3.

2.2. **Interval sums.** The general evaluation underlying the interval formulas is [21, (6)]: for a character χ modulo M, any multiple N of M, any r prime to N and any $m \geq 1$,

$$m\, r^{m-1} \sum \chi(n)\, n^{m-1} = -B_{m,\chi}\, r^{m-1} + \frac{\overline{\chi}(r)}{\varphi(r)} \sum_{\psi} \overline{\psi}(-N)\, B_{m,\chi\psi}(N). \tag{2}$$

Szmidt, Urbanowicz and Zagier trace this formula to Lerch [17], Berndt [1] and Urbanowicz [22]. In (2) the first sum is over $0 < n < N/r$ and the second is over characters ψ modulo r; $B_{m,\chi\psi}(X) = \sum_{0 \leq j \leq m} \binom{m}{j} B_{j,\chi\psi} X^{m-j}$ is the generalized Bernoulli polynomial of $\chi\psi$, regarded as a character modulo rN, evaluated at $X = N$. In [21] the formula is printed with $\chi(r)$ and $\psi(-N)$. That form agrees with (2) whenever $\chi(r)$ and every ψ modulo r are real, which covers every application in [21] (quadratic χ and $r \mid 24$). For non-real characters the conjugated form (2) is the correct one, and it is the form Girstmair applies to characters of order four and eight [10, (27)]. For instance, with $\chi = \chi_4$ normalized as in Definition 3.3, $p = 13$, $N = p$, $r = 2$ and $m = 1$, the left side of (2) is $1 + 3i$.

At $m = 1$, $N = p$ and $r = 2$, formula (2) for an odd character is the fixed-base identity (7) at $b = 2$: $\sum_{0<n<p/2} \chi(n) = (\overline{\chi}(2) - 2)B_{1,\chi}$. At $m = 2$ the formula gives the moment-weighted forms appearing in [1, §13]. The eighth- and sixteenth-range formulas in Berndt belong to the same linear theory as the trace above; what they do not supply is an ordering of the two conjugate quartic values.

Lean: the case $m = 1$, $N = p$, $r = 2$, as `half_sum`. The general formula (2) is quoted.

3. The coordinates and the pin

Proposition 3.1 (pairing and integrality)**.** *For $p \in \mathcal{P}^*$, the four S_j are integers, $S_0 + S_2 = S_1 + S_3 = (p-1)/4$, and*

$$k = S_0 - S_2, \qquad t = S_1 - S_3. \tag{3}$$

Proof. Put $m = (p-1)/4$. Since $p > 5$, $m > 1$, so H contains an element $a \neq 1$; multiplication by a permutes each coset C_j, so the sum of its elements is 0 in $\mathbb{F}_p$, and therefore the integer sum of the least positive representatives in C_j is divisible by p. Hence $S_j \in \mathbb{Z}$. Multiplication by a generator permutes the four cosets cyclically, so each has m elements. Since $p \equiv 5 \pmod 8$, $\chi_4(-1) = -1$, so $n \mapsto p-n$ interchanges C_j and C_{j+2}; pairing the m elements gives $S_j + S_{j+2} = m$. Finally $B_{1,\chi_4} = S_0 + iS_1 - S_2 - iS_3$, which gives (3). □

Lean: `p_dvd_cosetSum`, `card_class`, `S_add_S_neg`, `T_eq`.

Proposition 3.2 (oddness)**.** *For $p \in \mathcal{P}^*$, k and t are odd integers.*

Proof. By Proposition 3.1, $k = 2S_0 - (p-1)/4$ and $t = 2S_1 - (p-1)/4$. Since $(p-1)/4$ is odd, both coordinates are odd. □

Lean: `odd_k`, `odd_t`.

There are exactly two primitive quartic characters modulo p, χ_4 and $\overline{\chi_4}$. Conjugation fixes k and sends t to $-t$. By Proposition 2.1 the classical invariants already determine k and $|t|$, so the entire remaining ambiguity is the sign of the second coordinate.

Lean: `quartic_eq_or_eq_pow_three`, `k_pow_three`, `t_pow_three`.

Definition 3.3 (the pin)**.** For $p \in \mathcal{P}^*$, let χ_4 be the unique quartic character satisfying $\chi_4(2) = i$. This is possible and unique because $p \equiv 5 \pmod 8$ makes 2 a quadratic non-residue, so $\chi_4(2) \in \{i, -i\}$, and the two quartic characters take the two values.

Lean: `quartic_apply_two`, `existsUnique_pinned`.

Proposition 3.4 (the Jacobi-sum dictionary)**.** *Let χ_4 be pinned by $\chi_4(2) = i$ and write $J(\chi_4, \chi_4) = a + bi$. Then $a^2 + b^2 = p$, $a \equiv -1 \pmod 4$ and $b \equiv 2 \pmod 4$. Replacing χ_4 by $\overline{\chi_4}$ leaves a fixed and sends b to $-b$. Thus the pin on χ_4 and the sign choice for the second coordinate of the quartic Jacobi sum are the same conjugation choice.*

Proof. This is the standard quartic Jacobi-sum evaluation [19, Cor. 3.10]. The norm is $|J(\chi,\chi)|^2 = p$ for χ and χ^2 nontrivial. For the congruences, pair x with $1-x$ in $J = \sum \chi(x)\chi(1-x)$: the fixed point $x = 1/2$ contributes $\chi(1/4) = \chi(2)^{-2} = -1$, and the remaining terms pair off, so $J = 2W - 1$ where W is a sum of $(p-3)/2$ values in $\{\pm 1, \pm i\}$. Then $a = 2\operatorname{Re} W - 1$ and $b = 2\operatorname{Im} W$ with $\operatorname{Re} W + \operatorname{Im} W$ odd, and $a^2 + b^2 = p \equiv 5 \pmod 8$ forces $\operatorname{Im} W$ odd and $\operatorname{Re} W$ even. Complex conjugation commutes with the finite sum defining J, so $J(\overline{\chi_4}, \overline{\chi_4}) = a - bi$, and the pin chooses one of the two signs of b. □

Lean: `jacobiSum_norm`, `jacobiSum_mod_four`, `jacobiSum_pow_three`.

Remark (the sign selected by the pin)**.** If $\chi_4(2) = i$, then $b \equiv -2 \pmod 8$. For either quartic character χ, writing $J(\chi,\chi) = a + bi$, one has $\chi(2) = i^{ab/2}$: the Jacobi sum of the quartic residue character χ_π modulo a primary prime π of norm p satisfies $-\chi_\pi(-1)J(\chi_\pi,\chi_\pi) = \pi$, and the supplementary law gives $(2/\pi)_4 = i^{ab/2}$ for $\pi = a + bi$ [23, Ch. 9]. Here $\chi(-1) = -1$, and $a + bi$ is primary because $a \equiv 3 \pmod 4$ and $b \equiv 2 \pmod 4$. Hence $\chi_4(2) = i$ exactly when $ab/2 \equiv 1 \pmod 4$; since $a \equiv 3 \pmod 4$, this means $b \equiv -2 \pmod 8$. For $p = 13$, $J(\chi_4, \chi_4) = 3 - 2i$.

Lean: not formalized; the supplementary law for the quartic character of 2 is quoted.

Proposition 3.5 (quarter-count dictionary)**.** *Let $N_r = \#\{n \in H : rp/4 < n < (r+1)p/4\}$ for $0 \le r \le 3$, and put $A = N_0 - N_3$ and $B = N_1 - N_2$. Under the normalization $\chi_4(2) = i$, Hudson's formulas* [15, Thm. 3.1] *give*

$$\begin{aligned} S_0 &= \tfrac{1}{2}\big(\tfrac{p-1}{4} - \tfrac{3}{5}A - \tfrac{1}{5}B\big), & S_1 &= \tfrac{1}{2}\big(\tfrac{p-1}{4} - \tfrac{1}{5}A + \tfrac{3}{5}B\big), \\ S_2 &= \tfrac{1}{2}\big(\tfrac{p-1}{4} + \tfrac{3}{5}A + \tfrac{1}{5}B\big), & S_3 &= \tfrac{1}{2}\big(\tfrac{p-1}{4} + \tfrac{1}{5}A - \tfrac{3}{5}B\big). \end{aligned} \tag{4}$$

Consequently

$$k = -\frac{3A+B}{5}, \qquad t = \frac{-A+3B}{5}, \tag{5}$$

and conversely

$$A = -\frac{3k+t}{2}, \qquad B = \frac{3t-k}{2}. \tag{6}$$

Thus (k,t) and the two independent quarter-range differences are equivalent data. The four individual quarter counts are not determined by k and t alone.

Proof. Equation (4) is [15, Thm. 3.1] in the normalization $\chi_4(2) = i$. A short route to it: let Q_r be the sum of χ_4 over the r-th quarter of $(0,p)$. The fixed-base identity (7) at $b = 2$ and $b = 4$, with $\chi_4(2) = i$ and $\chi_4(4) = -1$, together with the reflection $Q_{3-r} = -Q_r$, gives $2Q_0 = (-3+i)\beta$ and $2Q_1 = (-1-3i)\beta$. Since $n \mapsto p-n$ carries $C_2 \cap$(quarter r) onto $H \cap$(quarter $3-r$), $\operatorname{Re} Q_0 = A$ and $\operatorname{Re} Q_1 = B$. Taking real parts gives (6); solving gives (5); and (5) with Proposition 3.1 gives (4). □

Lean: `two_A`, `two_B` for (6), `five_k`, `five_t` for (5), `hudson_S` for (4), cleared of denominators.

Theorem 3.6 (reconstruction)**.** *For $p \in \mathcal{P}^*$, the classical invariants determine the quartic value up to conjugation, and the pin determines it completely. More precisely, $k = \frac{1}{2}\operatorname{Tr}(\beta)$ and $|t| = \sqrt{2h^*(p) - k^2}$, and $\chi_4(2) = i$ selects the sign of t. With that choice, Proposition 3.4 fixes the corresponding sign of b in the Jacobi sum, and Proposition 3.5 reconstructs the quarter-range differences A and B:*

$$(\operatorname{Tr}\beta,\ h^*(p)) \longrightarrow (k, |t|) \xrightarrow{\chi_4(2)=i} (k,t) \longleftrightarrow (A,B).$$

Proof. The first two assertions are Proposition 2.1 and Definition 3.3. The two dictionaries are Propositions 3.4 and 3.5. □

Lean: `reconstruction` (every quartic ψ has $k(\psi) = k(\chi_4)$ and $t(\psi) = \pm t(\chi_4)$, $\psi(2) = i$ exactly when $\psi = \chi_4$, and $(k,t) \leftrightarrow (A,B)$); the step through $h^*(p)$ is quoted.

Example 3.7 ($p = 13$)**.** Here $H = \{1,3,9\}$, so $(N_0, N_1, N_2, N_3) = (2,0,1,0)$, hence $A = 2$ and $B = -1$. Formula (5) gives $k = t = -1$, so the pinned value is $\beta = -1 - i$. Its norm is 2, and Proposition 2.1 gives $h^*(13) = 1$. The example checks the integrality, parity, norm and quarter-count conventions together.

4. Fixed-base and diagonal families

The classical digit and coset-sum identities use a fixed base b. Multiplication by b is a bijection of $(\mathbb{Z}/p\mathbb{Z})^\times$, and composing a character with that map contributes a constant factor. For $b \ge 2$ with $p \nmid b$, put

$$\theta(b;n) = \frac{b\langle n\rangle - \langle bn\rangle}{p} = \left\lfloor \frac{bn}{p} \right\rfloor, \qquad 0 < n < p.$$

Proposition 4.1 (fixed-base collapse)**.** *For a character χ modulo p,*

$$\sum_{1\le n\le p-1} \overline{\chi}(n)\,\theta(b;n) = (b-\chi(b))\,B_{1,\overline{\chi}}. \tag{7}$$

Proof. Since $\theta(b;n) = (bn - \langle bn\rangle)/p$, the first term contributes $bB_{1,\overline{\chi}}$. In the second term the substitution $m = \langle bn\rangle$ is a bijection of $(\mathbb{Z}/p\mathbb{Z})^\times$ and $\overline{\chi}(b^{-1}m) = \chi(b)\overline{\chi}(m)$, so it contributes $\chi(b)B_{1,\overline{\chi}}$. Subtraction gives (7). □

Lean: `fixed_base_collapse`, for characters with values in any commutative ring; no parity hypothesis is needed.

This is the prime-conductor form of [11, Thm. 1]. The mechanism is the bijective constant-factor substitution. On the diagonal $\theta(n;n) = \lfloor n^2/p\rfloor$ the base moves with the summand: characters still satisfy $\chi(n^\ell) = \chi(n)^\ell$, but $n \mapsto n^\ell$ is $\gcd(\ell, p-1)$-to-one onto the ℓ-th powers, not a bijection carrying a constant character factor, so the proof of Proposition 4.1 does not apply to diagonal sums.

5. The higher 2-power levels

The coset description of Section 3 is the level-four case of a description at every 2-power level. In this section p is a prime with $v_2(p-1) = a \ge 2$, g is a primitive root modulo p, and χ is a character of exact order 2^a, with $\zeta = \chi(g)$ a primitive 2^a-th root of unity. Exactly as in Proposition 1.1, $\chi(-1) = \zeta^{(p-1)/2} = -1$, because $(p-1)/2^a$ is odd: χ is odd. For $b \le a$ let H_{2^b} be the subgroup of 2^b-th powers, of order $(p-1)/2^b$; the classes of χ are the cosets

$$C_r = \{n : \chi(n) = \zeta^r\} = g^r H_{2^a}, \qquad S_r = \frac{1}{p}\sum_{n\in C_r}\langle n\rangle \qquad (0 \le r < 2^a),$$

and $B_{1,\chi} = \sum_r \zeta^r S_r$. At $a = 2$ these are the C_j and S_j of Section 2.

Proposition 5.1 (the lower levels vanish)**.** *Let $b < a$. Every coset C of H_{2^b} is closed under $n \mapsto -n$, so $\frac{1}{p}\sum_{n\in C}\langle n\rangle = |C|/2$, and all coset differences at level 2^b vanish. Equivalently, $B_{1,\psi} = 0$ for every even character $\psi \ne 1$ modulo p.*

Proof. Since 2^b divides $(p-1)/2$, $-1 = g^{(p-1)/2}$ lies in H_{2^b}. Pairing n with $p-n$ in C gives $2\sum_C\langle n\rangle = |C|p$. For the character form, substitute $n \mapsto -n$ in $\sum\psi(n)\langle n\rangle$: it equals $\sum\psi(n)(p - \langle n\rangle)$, so $2\sum\psi(n)\langle n\rangle = p\sum\psi(n) = 0$. This is the classical vanishing of $B_{1,\psi}$ for even $\psi \ne 1$ [24, §4.1]. □

Lean: `levelSum_even` (coset form, for the classes of any even character), `even_T_eq_zero`.

So among the characters of order dividing 2^a only those of exact order 2^a have nonzero $B_{1,\chi}$, and only the top level carries coordinates.

Proposition 5.2 (integrality and parity at every level)**.** *Suppose that p is not a Fermat prime, that is, $|H_{2^a}| = (p-1)/2^a > 1$. Then the S_r are integers, $S_r + S_{r+2^{a-1}} = |H_{2^a}|$, and*

$$B_{1,\chi} = \sum_{0\le r<2^{a-1}} d_r\zeta^r, \qquad d_r = S_r - S_{r+2^{a-1}},$$

with every d_r an odd integer.

Proof. This is the proof of Propositions 3.1 and 3.2. An element $c \ne 1$ of H_{2^a} permutes each C_r, so the sum of C_r vanishes in $\mathbb{F}_p$ and $S_r \in \mathbb{Z}$. Since χ is odd, $n \mapsto p-n$ carries C_r onto $C_{r+2^{a-1}}$, which gives the pairing; and $\zeta^{r+2^{a-1}} = -\zeta^r$ gives the expansion. Finally $d_r = 2S_r - |H_{2^a}|$ with

$|H_{2^a}|$ odd: a subgroup of the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ of even order would contain its element of order two, which is -1, and $\chi(-1) = -1 \neq 1$. □

Lean: `p_dvd_levelSum`, `levelSum_add_neg`, `level_parity`; the oddness of $|H_{2^a}|$ is `card_kernel_odd`.

At a Fermat prime the subgroup is trivial and integrality fails: at $p = 17$, $a = 4$, one has $d_0 = 2 \cdot \frac{1}{17} - 1 = -\frac{15}{17}$. The integrality of coset sums of a nontrivial subgroup is classical [7, 11], and so is the parity at level four (Proposition 3.2); we have not found the parity at the levels $2^a \geq 8$ stated in print, and record it only because the argument is the level-four one verbatim.

Proposition 5.3 (the level invariant; Murty and Thangadurai)**.** *Let* $K_{2^a} = \frac{2}{p}\sum_{n \in H_{2^a}} \langle n \rangle - |H_{2^a}|$. *Then*

$$K_{2^a} = d_0 = 2^{1-a} \operatorname{Tr}_{\mathbb{Q}(\zeta_{2^a})/\mathbb{Q}} B_{1,\chi}.$$

In particular $K_4 = k$, *and* K_{2^a} *is an odd integer when* p *is not a Fermat prime.*

Proof. The conjugates of $B_{1,\chi}$ are the B_{1,χ^u} for odd u modulo 2^a, so the trace is $\frac{1}{p}\sum_n \langle n \rangle \sum_{u \text{ odd}} \chi(n)^u$. For $z = \chi(n)$ a 2^a-th root of unity, $\sum_{w<2^{a-1}} z^{2w+1} = z\sum_w (z^2)^w$ is 2^{a-1} if $z = 1$, -2^{a-1} if $z = -1$, and 0 otherwise. Hence the trace is $2^{a-1}(S_0 - S_{2^{a-1}}) = 2^{a-1}d_0$, and $d_0 = 2S_0 - |H_{2^a}| = K_{2^a}$ by the pairing. □

Lean: `level_trace`, in the form $\sum_{w<2^{a-1}} \sum_n \chi^{2w+1}(n)\langle n \rangle = 2^{a-1}(pS_0 - pS_{2^{a-1}})$; the identification of the sum over odd exponents with the field trace is the classical description of the Galois conjugates of $B_{1,\chi}$.

This is Theorem 1 of Murty and Thangadurai [18], there read as a statement about the digits of $1/p$, and restated at the 2-power levels by Girstmair [10, Thm. 1]. At $a = 2$ it is the trace in (1); at level two, for $p \equiv 3 \pmod 4$ and $p > 3$, the same coset sum is $K_2 = B_{1,\left(\frac{\cdot}{p}\right)} = -h(-p)$.

Example 5.4 (levels eight)**.** At $p = 41$ ($a = 3$), $H_8 = \{1, 10, 16, 18, 37\}$ has sum $82 = 2 \cdot 41$, so $K_8 = 2 \cdot 2 - 5 = -1$. At $p = 73$ ($a = 3$), $H_8 = \{1, 2, 4, 8, 16, 32, 37, 55, 64\}$ has sum $219 = 3 \cdot 73$, so $K_8 = 6 - 9 = -3$; with the primitive root $g = 5$ and $\chi(5) = \zeta_8$, the coordinates are $(d_0, d_1, d_2, d_3) = (-3, -3, 3, 1)$, all odd.

Beyond these facts the higher levels enter the eighth- and sixteenth-range sums through (2) at $r = 8$ and $r = 16$, together with Bernoulli numbers of twists of conductor $8p$ and $16p$ [1, §10], [10]; as at level four, these linear formulas do not order the conjugates.

6. Attribution register

The register records the classical inputs used above.

statement	source
quadratic excess on $(0, p/2)$ and $(0, p/4)$	Dirichlet 1839; see [1, (1.1), (1.2)]
interval sums as linear combinations of $L(1, \chi\psi)$, every modulus and moment	[21, (6)], after Lerch [17], Berndt [1], Urbanowicz [22]; conjugated form as in [10, (27)]
eighth- and sixteenth-range character sums	[1, §7, §10]
moment-weighted interval sums	[1, §13]; equation (2) at $m \geq 2$
relative class number as a group determinant in coset sums, through degree ten	[4], restated in [16]
relative class number of the prime-conductor quartic field in coset sums	Hudson–Williams [14]; proof in Hudson [15, §2]
quarter-count / coset-sum conversion in the quartic case	Hudson [15, Thm. 3.1]
quartic Jacobi-sum normalization $a^2+b^2 = p$, $a \equiv -1 \pmod 4$	[19, Cor. 3.10]
sign of b selected by the pin: $b \equiv -2 \pmod 8$, i.e. $\chi(2) = i^{ab/2}$	supplementary law for 2; [23, Ch. 9]
coset-sum family and deviation vector; norm-type L-value product	[7], surveyed in [11]
digits of $1/p$ and $h(-p)$	[5]; [6, Satz 11]
trace of first generalized Bernoulli numbers in the digit formula	[18, Thm. 1]; [10, Thm. 1]
$B_{1,\psi} = 0$ for even $\psi \neq 1$	[24, §4.1]
integrality of coset sums of a nontrivial subgroup	[7], [11]
parity of the coordinates at levels $2^a \geq 8$	argument of Proposition 3.2; not found in print
fixed-base twisted digit identity (7)	[11, Thm. 1]
digit frequencies as class numbers and classification of when this is possible	[9]; [12]
digit variance and Dedekind sums	[8]; [13]; [10]
$p \equiv 1 \pmod 4$ digit symmetry via composite discriminant	[2]
quadratic residue counts in arithmetic progressions via the class-number formula	[3]
absence of bias in the function-field analogue	[20]

7. Formalization

The accompanying Lean 4 development `QuarticSurvey` (Lean `v4.35.0`, Mathlib `v4.35.0`; https://github.com/jkfarland-bot/KSurvey, release v1.1, archived on Zenodo, all versions doi:10.5281/zenodo.23624362) works with characters $\chi : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}[i]$ of order four, written `MulChar (ZMod p) GaussianInt`, and, for Section 5, with complex-valued characters `MulChar (ZMod p)` $\mathbb{C}$, and with sums over least nonnegative residues. It contains no `sorry` and no `native_decide`, and every result below depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The hypotheses are those of the text: p prime, $p \equiv 5 \pmod 8$, and $p > 5$ where the coset sums enter. In Section 5 the hypotheses are stated without a: $\chi(-1) = -1$ for oddness, and $\chi^m = 1$ for some $0 < m < p-1$ in place of "p is not a Fermat prime" (this gives a nontrivial kernel).

statement	Lean	file
Prop. 1.1, odd orders are even	`apply_neg_one_of_odd_orderOf`	Basic
Prop. 1.1, order two is real	`apply_of_orderOf_two`	Basic
Prop. 1.1, order four: odd iff $p \equiv 5 \pmod 8$	`quartic_apply_neg_one`, `quartic_apply_neg_one_of_one`	Basic
$\chi_4(2) = \pm i$	`quartic_apply_two`	Basic
Def. 3.3, the pin exists and is unique	`existsUnique_pinned`	Pin
the two quartic characters are χ_4 and $\overline{\chi_4}$	`quartic_eq_or_eq_pow_three`	Pin
Prop. 3.1, integrality	`p_dvd_cosetSum`	CosetSums
Prop. 3.1, class sizes $(p-1)/4$	`card_class`	CosetSums
Prop. 3.1, pairing $S_j + S_{j+2} = (p-1)/4$	`S_add_S_neg`	CosetSums
Prop. 3.1, $\beta = k + it$	`T_eq`	CosetSums
Prop. 3.2, oddness	`odd_k`, `odd_t`	CosetSums
conjugation fixes k, negates t	`k_pow_three`, `t_pow_three`	CosetSums
Prop. 2.1, trace	`trace_eq`	Reconstruction
Prop. 3.4, $a^2 + b^2 = p$	`jacobiSum_norm`	Jacobi
Prop. 3.4, $a \equiv -1$, $b \equiv 2 \pmod 4$	`jacobiSum_mod_four`	Jacobi
Prop. 3.4, $J(\overline{\chi}, \overline{\chi}) = \overline{J(\chi, \chi)}$	`jacobiSum_pow_three`	Jacobi
Prop. 3.5, (6), (5), (4)	`two_A`, `two_B`, `five_k`, `five_t`, `hudson_S`	Quarters
Thm. 3.6, formal part	`reconstruction`	Reconstruction
Prop. 4.1	`fixed_base_collapse`	FixedBase
(2) at $m = 1$, $N = p$, $r = 2$	`half_sum`	FixedBase
Prop. 5.1, character and coset forms	`even_T_eq_zero`, `levelSum_even`	Levels
Prop. 5.2, integrality and pairing	`p_dvd_levelSum`, `levelSum_add_neg`	Levels
Prop. 5.2, parity	`level_parity`, `card_kernel_odd`	Levels
Prop. 5.3	`level_trace`	Levels

Quoted and not formalized: the identification $\mathrm{N}(\beta) = 2h^*(p)$ in Proposition 2.1, which needs relative class numbers of number fields; the L-value normalization of Section 2; the general formula (2); the Remark $b \equiv -2 \pmod 8$, which needs the quartic character of 2; the identification of the sum over odd exponents with the field trace in Proposition 5.3; and the numerical Examples 3.7 and 5.4, which are checked by direct computation.

8. Questions

These record gaps in the literature as found, not claims.

Question 8.1. *Berndt's pairing at the sixteenth range* [1, §10] *and the pairing that appears in coordinate congruences at level eight both arise when characters of order greater than two enter. Is there a uniform statement of this pairing across the* 2*-power levels?*

Question 8.2. *Does the quartic-order Bernoulli number in* [9, Rem. 5]*, of conductor* $5p$*, admit a coordinate description analogous to the prime-conductor case: a Galois-invariant coordinate determined before pinning and a complementary coordinate with a natural arithmetic normalization?*

Question 8.3. *Berndt* [1, §14] *reports, as a written communication from Elliott, that for characters* χ_p *of fixed order in a residue class the values* $\arg L(1, \chi_p)$ *are everywhere dense modulo* 2π*. What is the published form of that result?*

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